Supplemental Material: Finite-length and vertical effects in nonlinear-kicker injection into a Korea Light Source design lattice

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TABLE S1. Independent trajectory check. Launch y is in mm, y' in mrad and δ in 10^{-3}; absolute coordinate discrepancies are nm for positions and nrad for canonical momenta. Deterministic solver comparisons, without sampling intervals.

Case	y	y'	δ	$\lvert\Delta x\rvert$	$\lvert\Delta p_x\rvert$	$\lvert\Delta y\rvert$	$\lvert\Delta p_y\rvert$
nominal	0.0	0.0	0.0	0.045	0.525	0.000	0.000
lower angle	0.0	0.0	0.0	0.032	0.010	0.000	0.000
upper angle	0.0	0.0	0.0	0.048	0.737	0.000	0.000
vertical position	1.0	0.0	0.0	0.049	0.447	0.027	0.814
vertical angle	0.0	0.2	0.0	0.045	0.525	0.004	0.007
both vertical	1.0	0.2	0.0	0.048	0.449	0.031	0.823
$\delta < 0$	1.0	0.2	-2.0	0.048	0.449	0.031	0.824
$\delta > 0$	1.0	0.2	2.0	0.048	0.449	0.031	0.821

This supplement collects detailed tables, a larger-emittance sensitivity case and procedural definitions that support the main article. Numbers are generated from the same evidence manifest as the article.

S1. INDEPENDENT TRAJECTORY CHECK

An adaptive DOP853 integration of the stated paraxial equations, in the same filament field, is compared with the 640-slice drift–kick–drift map at the nominal point, at the interpolated angular-window edges (offsets -0.319/0.124 mrad), with nonzero vertical position and angle, and at $\delta = \pm 0.002$. Relative/absolute solver tolerances are $2 \times 10^{-11}/2 \times 10^{-13}$. Table S1 lists the discrepancies; slice and span checks through 1280 slices and ± 0.9 m stay below one percent of a 0.1σ response in either plane.

S2. WIRE-POSITION AND LOOP-CURRENT SENSITIVITIES

Tables S2 and S3 give the first-order on-axis dipole and quadrupole sensitivities of the ideal geometry to each wire coordinate and to each loop current, and the single-error limits for a stored-beam centroid amplitude of 0.1σ in each plane, at the selected field scale $S = 0.95$

TABLE S2. On-axis wire sensitivity. Wire numbers follow the four quadrant loops, inner then outer wire. u is the displaced coordinate. Dipole derivatives $D_{x,y} = \partial K_{x,y}/\partial u$ are in rad/m; $Q_n = \partial J_{xx}/\partial u$ and $Q_s = \partial J_{xy}/\partial u$ are in rad/m^2, with $J = \partial(K_x, K_y)/\partial(x, y)$, $J_{yy} = -J_{xx}$ and $J_{yx} = J_{xy}$ to numerical precision. t_x, t_y are single-error 0.1σ limits in μm. A dash indicates no resolved first-order dipole sensitivity; higher orders remain. Deterministic central differences, step $0.1\,\mu$m.

Wire	u	D_x	D_y	Q_n	Q_s	t_x	t_y
1	x	0	+0.399	-75.3	+75.3	–	0.26
1	y	+0.399	0	+75.3	+75.3	0.49	–
2	x	0	-0.107	+10.4	-10.4	–	0.98
2	y	-0.107	0	-10.4	-10.4	1.82	–
3	x	0	-0.399	+75.3	+75.3	–	0.26
3	y	-0.399	0	+75.3	-75.3	0.49	–
4	x	0	+0.107	-10.4	-10.4	–	0.98
4	y	+0.107	0	-10.4	+10.4	1.82	–
5	x	0	+0.399	+75.3	-75.3	–	0.26
5	y	+0.399	0	-75.3	-75.3	0.49	–
6	x	0	-0.107	-10.4	+10.4	–	0.98
6	y	-0.107	0	+10.4	+10.4	1.82	–
7	x	0	-0.399	-75.3	-75.3	–	0.26
7	y	-0.399	0	-75.3	+75.3	0.49	–
8	x	0	+0.107	+10.4	+10.4	–	0.98
8	y	+0.107	0	+10.4	-10.4	1.82	–

(the derivatives scale linearly with S). The main article reports the simultaneous-error consequence.

S3. LARGER-EMITTANCE SENSITIVITY CASE

Before the design booster beam was supplied, the studies used a 100/10 nm rad beam with rms energy spread 0.0011, the septum aperture and no additional chamber profile. For

TABLE S3. Independent-loop relative-current sensitivity. Dipole derivatives are in rad per relative current; normal and skew quadrupole derivatives in rad/m per relative current. The final column is the joint two-plane 0.1σ single-loop limit in units of 10^{-4}. Deterministic derivatives with relative-current step 10^{-5}.

Loop	D_x	D_y	Q_n	Q_s	Limit
1	-0.0010	-0.0010	+0.000	-0.292	1.03
2	+0.0010	-0.0010	+0.000	+0.292	1.03
3	+0.0010	+0.0010	+0.000	-0.292	1.03
4	-0.0010	+0.0010	+0.000	+0.292	1.03

1000 particles (seeds 1001, 1002), 93.0 % survive 500 turns and 89.5 % survive 4000 turns (95 % particle Wilson interval 87.4–91.3 %); the late loss is 3.5 percentage points (Fig. S1). A separate 2000-particle sample gives 92.4 % (91.2–93.5 %) at 500 turns. In that case the paired vertical-kick diagnostic gives 93.0 % with the NKM vertical kick and 94.8 % when it is suppressed; a uniform 5 mm vertical half-aperture reduces capture with or without that kick (to 86.3 and 88.0 %), which supports a role for transverse transfer in the ring without identifying a unique coupling resonance. These numbers belong to the larger-emittance case and are not design-beam results.

S4. STATISTICAL METHODS

Particle intervals. Capture of a fixed bunch is a binomial fraction; its 95 % interval is the Wilson score interval over the pooled particles of the stated seeds. It describes particle sampling for fixed models, ring and settings only.

Realization intervals. Joint machine/shot ensembles are summarized by their mean, median, tenth percentile and fraction of realizations at or above 80 %. Intervals of the mean are percentile bootstraps over joint realizations (10 000 resamples, seed 1729). Because each realization draws one of ten corrected rings with replacement, a second, ring-cluster bootstrap resamples the ten ring identities and weights their realizations; together with the leave-one-ring-out means it shows how much the result depends on the finite ring population.

Paired differences. Comparisons between models, jitter levels or tuned and untuned

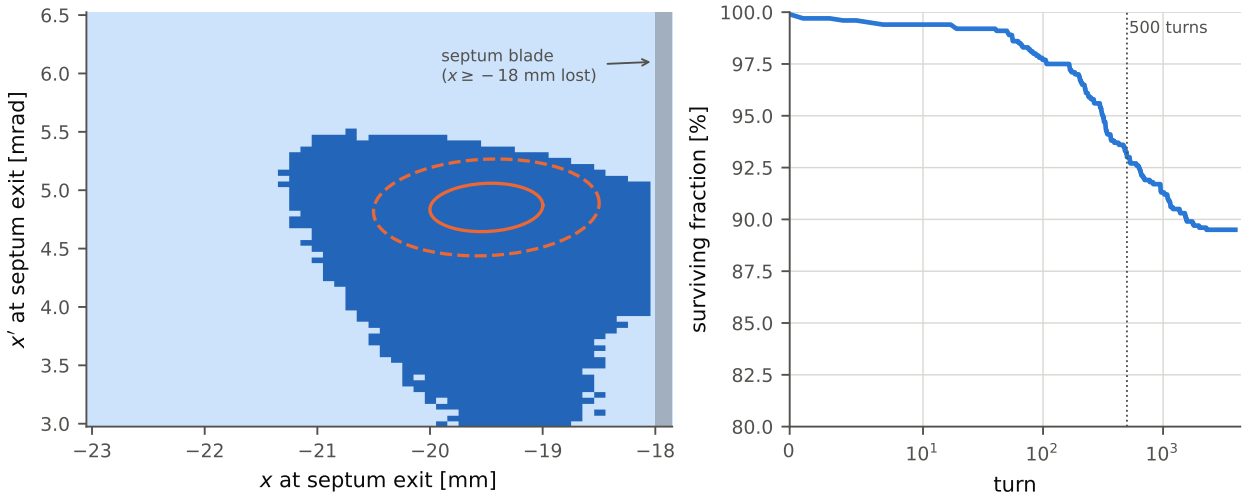


FIG. S1. Larger-emittance sensitivity case without the additional chamber profile. Left: deterministic 500-turn single-particle survival in septum phase space ($\delta = 0$, 3692 launches), with 100/10 nm rad beam ellipses at one and two rms widths. Right: survival of 1000 particles, seeds 1001, 1002, through 4000 native turns; the final 95 % Wilson interval is stated in the text.

settings use the same particles and error draws; the paired difference is bootstrapped over realizations (10 000 resamples, seed 1729).

Acceptable-loss test. Whether a jitter level costs less than one percentage point is a one-sided noninferiority question: the lower endpoint of the central 95 % bootstrap interval of the paired difference must lie above −1 percentage point. That endpoint is a conservative 97.5 % one-sided bound. The one-point value is the declared analysis margin of the robustness analysis; it is not a design specification, and an interval that crosses it is reported as inconclusive rather than as equivalence.

For the two-plane scan, the paired acceptable-loss comparison is specifically NKM(scale) minus NKM(zero), with all non-jitter draws fixed. It is distinct from NKM minus the uniform kick. The uniform-kick cost and the change of that model gap use the same pairing (table S4). Intervals are pointwise across five sampled scales, without a simultaneous-band or continuous-threshold interpretation. The zero-jitter row is a reference and cannot demonstrate a nonzero tolerance.

Associations. The round-4 analysis enters signed errors with their signs (booster position, angle and energy, NKM-to-orbit offset, NKM scale, septum position and strength, β mismatch) and a physically constructed predictor: the horizontal betatron amplitude of the

TABLE S4. Additional two-plane jitter summaries for the main article's paired scan: 50 realizations of 200 particles at 500 turns, 10 000 realization-bootstrap resamples, seed 1729. Zero-capture fractions are percentages of realizations; changes are percentage points with paired central 95 % intervals. Gap denotes NKM minus uniform-kick capture.

Scale	NKM zero [%]	Uniform zero [%]	Uniform change from zero	Gap change from zero
0	0	0	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
0.1	0	0	+0.2 [-0.2, +0.7]	-2.6 [-4.6, -0.8]
0.25	0	0	-1.7 [-3.1, -0.5]	-9.7 [-14.4, -5.4]
0.5	8	0	-8.3 [-13.8, -3.8]	-16.7 [-23.5, -10.2]
1	28	10	-25.4 [-34.8, -16.6]	-17.2 [-25.4, -9.2]

centroid just after each kicker, computed from each realization's errors with the drawn ring's Twiss at the NKM. Spearman rank correlations with the NKM capture, the dipole capture and their paired difference over the 150 full-budget realizations carry two-sided asymptotic probabilities, Holm-adjusted within each response over the eleven predictors. They describe associations within the sampled model, not causal error rankings; the attribution uses the paired ablation of the main article.

Ablation. Each error class is disabled by setting its standard deviations to zero. Its draws are still taken from the same random streams, so every other error, the particle seeds and the ring identities are unchanged. Alone and omitted effects, and the non-additivity (full minus all-off, minus the sum of alone effects), are bootstrapped over the 50 paired realizations.

S5. PROCEDURES

Corrected rings. Each of ten error draws (seeds 5501–5510) is commissioned in ten steps of increasing error amplitude. BPM readings are referenced to the adjacent quadrupole centre with a 10 μm rms residual; the orbit is corrected with the 120 corrector pairs by truncated SVD (cutoff 10^{-3} of the largest singular value, ± 1 mrad bound) until the rms improves by less than 1 % per iteration. The β-beat and horizontal dispersion at the BPMs are then fitted with the 344 individual quadrupole gradients (Gauss–Newton on a fixed forward-difference Jacobian, SVD cutoff 10^{-2}, changes bounded to 1 % per step, three accepted steps), followed

by bounded Q11/Q12 tune correction and a final orbit correction. No skew correctors exist; coupling is not corrected.

Operating point. Candidates place x_{inj} at the $3\sigma_x$ septum-clearance limit and 0.25, 0.5 and 1.0 mm further out, with $S = 0.80$–1.00 and the inner matched root of the 3-D magnet; each is scored by the tenth percentile, then the mean, of capture over the corrected rings (100 particles per ring, no injection errors), for 2 mm and 1 mm blades. The selected points are validated on fresh 2×500-particle bunches; the paired dipole control is calibrated to the 3-D centroid response at each validated point.

Window. Capture is evaluated on a fixed 0.2 mm/mrad grid of septum-exit offsets about the selected point, extended while a boundary remains above 80 % (maximum offset 3), with crossing cells refined once to 0.1 and linearly interpolated. Two 500-particle bunches (seeds 1001/1002) per value.

Tuning. For each of 50 static machines, capture of 100 fresh training particles (all shot errors zero) is maximized over x_{inj} (bounded above by the clearance limit, 1.5 mm range), an angle trim of ±0.5 mrad about the re-matched angle of the current (x_{inj}, S), and S (0.75–1.05), by repeated strict-improvement coordinate sweeps (steps 0.25 mm, 0.1 mrad, 0.025; at most four sweeps). Held-out validation restores the joint shot errors on the paired 200-particle samples at zero and quarter booster jitter. The fraction of machines whose optimum lies on a bound is reported.